

Near-Perfect Session Retrieval on LongMemEval:

MemoryKG’s Hybrid Semantic-Graph Architecture
Achieves 99.4% Recall@10 and 100% Recall@30 Across 500 Questions

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Abstract

We evaluate MemoryKG, a conversational memory system based on DocKG [Suchanek, 2026b], on the LongMemEval benchmark [Wang et al., 2024], a dataset of 500 questions spanning six question types that require locating specific sessions within a multi-session conversation corpus of up to 500 sessions. MemoryKG achieves **99.4% session-level Recall@10** and **100% Recall@30** across all question types, with an NDCG@10 of 0.943, on the full 500-question test set. Perfect recall is attained at $k=30$ with no LLM required at any stage — neither for indexing nor for retrieval. The hardest question types are *temporal-reasoning* (Recall@10 = 98.5%) and *multi-session* (Recall@10 = 99.2%), where evidence spans multiple chronologically ordered sessions. The easiest types — *single-session-user*, *single-session-assistant*, and *single-session-preference* — each reach 100% Recall@10. Graph expansion through structured CONTAINS, NEXT, and REFERENCES edges closes the gap between semantic seeding and complete evidence recovery, requiring only $k=30$ seeds to exhaust all 500 questions. A single shared sentence-transformer embedder (BAAI/bge-small-en-v1.5) processes each per-question corpus in approximately 0.36 seconds on Apple Silicon.

1 Introduction

Long-term conversational memory retrieval — locating the right session from a large multi-session corpus given a natural-language question — is increasingly important for AI assistants operating over months of interaction history. Prior approaches include full-context LLM inference, structured memory extraction [Mem0, 2024], and flat dense retrieval over session text. These approaches differ in accuracy, latency, cost, and privacy guarantees.

The LongMemEval benchmark [Wang et al., 2024] provides a structured evaluation across six question types, each requiring the system to locate the correct session(s) from a corpus of up to 500 long-horizon conversation sessions. The benchmark distinguishes single-session retrieval (facts, preferences, assistant claims) from multi-session and temporally-ordered evidence.

MemoryKG [Suchanek, 2026a] builds a lightweight knowledge graph over verbatim conversation text, where semantic vector search seeds initial candidates and structured graph traversal expands the result set. MemoryKG is a direct specialisation of DocKG [Suchanek, 2026b], a hybrid semantic-graph system for general document corpora. In this paper we report MemoryKG’s performance on LongMemEval and analyse the retrieval characteristics of each question type.

2 The LongMemEval Benchmark

LongMemEval [Wang et al., 2024] is a benchmark for evaluating chat assistants on long-term interactive memory. It contains 500 questions drawn from realistic multi-session conversation

histories. Each question is paired with one or more *evidence sessions* — the specific conversation turns or sessions whose content is necessary to answer the question correctly.

The six question types are:

Single-Session User ($n=70$) Questions about facts the user stated in a single session.

Single-Session Assistant ($n=56$) Questions about claims or information the assistant provided in a single session.

Single-Session Preference ($n=30$) Questions about user preferences expressed within a single session.

Knowledge Update ($n=78$) Facts that the user updated or corrected across sessions; the system must retrieve the most recent statement.

Multi-Session ($n=133$) Questions whose answer requires aggregating information from two or more separate sessions.

Temporal Reasoning ($n=133$) Questions that require reasoning about the order or timing of events across sessions.

We use *session-level retrieval recall* as our primary metric: the fraction of evidence sessions whose text is found in the top- k retrieved nodes. We also report NDCG@ k (Normalised Discounted Cumulative Gain), which rewards placing the correct session higher in the ranked list. These are binary retrieval metrics, not end-to-end QA accuracy. The LongMemEval paper reports end-to-end QA accuracy (requiring an LLM to generate an answer); we report whether the evidence session is present in the retrieved context. These are complementary but distinct measures.

3 MemoryKG Architecture

MemoryKG represents a conversation corpus as a four-layer knowledge graph:

1. **Document layer.** Each conversation session is a root document node containing the full session text.
2. **Section layer.** Markdown headings (one per speaker turn) produce section nodes connected to their parent document by CONTAINS edges.
3. **Chunk layer.** Section content is split into sentence-aligned chunks; adjacent chunks are linked by NEXT edges.
4. **Semantic index.** All nodes are embedded with a sentence transformer and stored in LanceDB for approximate nearest-neighbour retrieval.

3.1 Query pipeline

A query q is processed in two stages:

Stage 1 — Semantic seeding. The top- k nodes by cosine similarity to the query embedding are retrieved from LanceDB. These form the *seed set* S .

Stage 2 — Graph expansion. Starting from S , BFS expansion walks outward through the graph for h hops along any edge in the relation set $\mathcal{R}$. The default relation set is

$\mathcal{R} = \{\text{CONTAINS, NEXT, REFERENCES, SIMILAR_TO, HAS_TOPIC, MENTIONS_ENTITY, HAS_KEYWORD}\}$

Nodes reachable within h hops of any seed are added to the candidate pool and ranked by a composite key combining seed distance, hop depth, and structural position.

For LongMemEval evaluations we use $k=10$ seeds and $h=1$ hop.

3.2 Why graph expansion helps for long-horizon retrieval

Consider a Knowledge Update question: “What is the user’s current job title?” The earliest session might read “I just started as a junior analyst,” while a later session reads “Got promoted to senior analyst last month.” A flat vector search that ranks the earlier session higher will return a stale answer. Graph expansion helps via CONTAINS edges: the parent document node for each session contains the full session text, and temporal ordering is encoded in NEXT edges between sessions. When the correct (most recent) session is seeded by any related chunk, expansion propagates recall upward to the session level.

For temporal reasoning questions, the same mechanism resolves ordering ambiguity: if any chunk from the target session appears in the seed set, traversal lifts the entire session into the candidate pool regardless of whether the question vocabulary matches the session’s surface text.

4 Experimental Setup

4.1 Corpus format

For each question, all sessions in the user’s history are written to a temporary directory as Markdown files (one file per session). Each speaker turn becomes a ##-headed section:

```
# Session 2024-03-15

## User

I just got promoted to senior analyst last month.

## Assistant

Congratulations! That’s a big step --- what changed in your role?

## User

More client-facing work, mostly. I’m leading the quarterly reviews now.
```

The heading chunk strategy splits at ## boundaries, so each turn becomes one chunk node.

4.2 Build configuration

- Chunk strategy: `heading` (no per-file embedding at parse time)
- Topic / entity / keyword enrichment: disabled (speed)
- SIMILAR_TO edge discovery: disabled (speed)
- Workers: 1 (per-item builds are already fast)
- Batch size: 512 embeddings

4.3 Shared embedder

A single `SentenceTransformerEmbedder` (BAAI/bge-small-en-v1.5, 384-dimensional) is initialised once and reused across all questions. This avoids the per-question model reload overhead

and is the primary contributor to the 0.36 s/question throughput. The full 500-question evaluation completes in 181.6 seconds on an Apple M5 Max.

4.4 Recall and NDCG computation

For each question, let $E = \{e_1, \dots, e_n\}$ be the set of evidence session texts and $N = \{n_1, \dots, n_m\}$ be the ranked list of retrieved node texts. Session-level recall at k is:

$$\text{Recall@}k = \frac{1}{|E|} \sum_{e \in E} \mathbf{1}[\exists n \in N_k : e \subset n \vee n \subset e]$$

where $\subset$ denotes case-insensitive substring containment and N_k is the top- k subset of N . $\text{NDCG@}k$ is computed over binary relevance labels using the standard formulation:

$$\text{NDCG@}k = \frac{\text{DCG@}k}{\text{IDCG@}k}, \quad \text{DCG@}k = \sum_{i=1}^k \frac{\text{rel}_i}{\log_2(i+1)}$$

where $\text{rel}_i \in \{0, 1\}$ indicates whether position i in the ranking contains an evidence session.

5 Results

5.1 Session-level retrieval across k

Table 1 shows MemoryKG’s session-level $\text{Recall@}k$ and $\text{NDCG@}k$ across increasing retrieval depths.

Table 1: MemoryKG session-level retrieval metrics on LongMemEval (500 questions).

k	Recall@k	NDCG@k
1	0.904	0.904
3	0.974	0.942
5	0.984	0.942
10	0.994	0.943
30	1.000	0.942
50	1.000	0.942

Perfect recall is reached at $k=30$: all 500 evidence sessions appear in the top-30 retrieved nodes. The $\text{NDCG@}k$ stabilises at 0.942–0.943 for $k \geq 3$, indicating that the correct sessions consistently rank near the top of the list once they appear.

5.2 Per-type breakdown

Table 2 shows $\text{Recall@}5$, $\text{Recall@}10$, and $\text{NDCG@}10$ for each LongMemEval question type.

Table 2: MemoryKG retrieval metrics on LongMemEval by question type (top-10, hop=1).

Question Type	n	Recall@5	Recall@10	NDCG@10
knowledge-update	78	1.000	1.000	0.984
multi-session	133	0.992	0.992	0.949
single-session-assistant	56	1.000	1.000	1.000
single-session-preference	30	1.000	1.000	0.913
single-session-user	70	0.986	1.000	0.932
temporal-reasoning	133	0.955	0.985	0.901
Overall	500	0.984	0.994	0.943

Four of the six question types achieve 100% Recall@10. The two types that do not — *temporal-reasoning* (98.5%) and *multi-session* (99.2%) — are those requiring evidence from multiple or temporally ordered sessions.

5.3 Comparison with literature baselines

Table 3 places MemoryKG in the context of systems evaluated in the LongMemEval literature [Wang et al., 2024]. Those results are reported as end-to-end QA accuracy, which requires an LLM to generate a free-form answer given retrieved context. MemoryKG’s retrieval recall is a strictly easier task: it measures whether the evidence is *present* in the retrieved context, not whether a reader can extract the correct answer from it. The two metrics are complementary but not directly comparable; the table is included for broader orientation.

Table 3: Context comparison. † QA accuracy from Wang et al. [2024] (requires LLM answer generation); ours is retrieval recall (no LLM). Different metrics; not directly comparable.

System	Score	Metric	LLM required
MemoryKG (ours, R@10)	99.4%	Retrieval recall	None
MemoryKG (ours, R@30)	100%	Retrieval recall	None
GPT-4o full-context†	~66%	QA accuracy	Yes
GPT-4o + RAG†	~50%	QA accuracy	Yes
GPT-4-Turbo full-context†	~56%	QA accuracy	Yes
Claude-3.5-Sonnet†	~52%	QA accuracy	Yes
ReadAgent†	~42%	QA accuracy	Yes
Zero-shot (no memory)†	~25%	QA accuracy	Yes

5.4 Miss analysis

At $k=10$, exactly 3 out of 500 questions returned no evidence session in the top-10 retrieved nodes. These three misses correspond to question IDs 10d9b85a, gpt4_468eb064, and eac54add. All three were recovered by $k=30$, confirming that the evidence exists in the index but ranks below position 10 in these cases. The miss rate at $k=10$ is 0.6% (3/500).

6 Analysis

6.1 Why temporal-reasoning is the hardest type

Temporal-reasoning questions ask about the ordering or recency of events across sessions (e.g., “When did the user first mention they were learning Spanish?”). The evidence session may contain

only a passing reference to the target event, while semantically closer sessions discuss unrelated topics. Vocabulary mismatch between the question and the target session is highest in this category: a question about “first mention” may correspond to a session that simply reads “I started a Spanish class” without temporal markers. Graph expansion partially compensates — if any chunk from the same session appears in the seed set, NEXT and CONTAINS edges recover the full session — but 2 out of 133 temporal-reasoning questions (1.5%) still miss at $k=10$.

6.2 Why multi-session questions remain hard

Multi-session questions require evidence from two or more sessions simultaneously. At $k=10$, the system must place all required sessions within the top-10 result, which becomes more constrained as the number of required sessions grows. Unlike the ConvoMem tier model [Pakhomov et al., 2025] (which fixes the number of evidence messages), LongMemEval multi-session questions vary in evidence count. The 1 miss at $k=10$ for this category represents a case where one required session ranked below position 10 while the other(s) ranked highly.

6.3 The NDCG plateau and its implications

$\text{NDCG}@k$ rises sharply from 0.904 at $k=1$ to 0.942 at $k=3$ and then plateaus at 0.942–0.943 for all larger k . This plateau indicates that additional retrieved nodes beyond $k=3$ do not substantially improve the rank of correct sessions — the correct sessions are already near the top of the list when they are present. The gap between $\text{Recall}@k$ and the NDCG plateau reflects the 3 hard misses that only appear beyond position 10. For a downstream reader consuming the top- k retrieved context, $k=10$ is effectively optimal: additional items improve coverage without meaningfully improving precision.

6.4 Throughput and scaling

At 0.36 s/question on Apple Silicon (M5 Max), MemoryKG processes 500 questions in 181.6 seconds. The dominant cost is LanceDB indexing (Phase 2) per question, which scales with the number of sessions in the corpus. LongMemEval corpora are larger than ConvoMem corpora (up to 500 sessions vs. up to 300 conversations), accounting for the higher per-question time (0.36 s vs. 0.07 s on ConvoMem). The shared embedder eliminates model reload overhead; the remaining cost is proportional to corpus size and is bounded by MPS-accelerated LanceDB ingestion.

7 Conclusion

MemoryKG achieves 99.4% $\text{Recall}@10$ and 100% $\text{Recall}@30$ on the 500-question LongMemEval benchmark, with an $\text{NDCG}@10$ of 0.943 and no LLM required at any stage. Four of six question types reach perfect $\text{Recall}@10$. The two remaining types — temporal-reasoning and multi-session — reflect the fundamental difficulty of retrieving evidence from multi-session contexts with temporal ordering requirements; both approach but do not reach 100% at $k=10$, and both are fully resolved by $k=30$.

The hybrid architecture — semantic seeding combined with structured graph expansion — enables high-precision retrieval without the latency or cost of LLM inference. The NDCG plateau at 0.942 from $k=3$ onward confirms that when evidence sessions are retrieved, they rank near the top of the result list, making the system suitable for downstream readers with tight context budgets.

Limitations. Recall measures whether the evidence session is *present* in the retrieved context; it does not measure whether a downstream reader can correctly extract the answer. End-to-end

QA accuracy on LongMemEval with a MemoryKG retrieval front-end is left for future work. The 3 misses at $k=10$ are all resolved by $k=30$ and are concentrated in the temporal-reasoning category, suggesting that temporal ordering cues are the primary remaining failure mode.

Reproducibility. All benchmark scripts, result JSON files, and this writeup are included in the MemoryKG repository. The full evaluation can be reproduced with:

```
python benchmarks/longmemeval/longmemeval_bench.py --limit 500
```

References

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